TEK ZANITH

Senior QA Automation Engineer | AI & Test Architecture Lead

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AI-native Senior Quality Assurance Engineer and Test Architect with 4+ years of experience leading end-to-end quality engineering across distributed web, mobile native, and enterprise systems. Specialized in **multi-agent test orchestration**, **Playwright & Maestro test automation**, and **code-level AST defect isolation** across high-concurrency fintech, e-government, and B2B SaaS platforms. Proven track record eliminating release bottlenecks by replacing manual testing with autonomous swarms, multi-device mobile validation fleets, and zero-leak Row Level Security (RLS) enforcement.

CORE COMPETENCIES & TECHNICAL STACK

TEST AUTOMATION & ENGINEERING

Playwright Automation, Maestro CLI, Selenium WebDriver (Python/C#), Bun Test Runner, Postman API Testing, Regression Suites, E2E Workflow Architecture.

AI & AGENTIC ORCHESTRATION

Multi-Agent Testing Swarms, Model Routing (Opus, Sonnet, Codex), Prompt Engineering, Deterministic Mocking, Synthetic Data, AST Bug Isolation.

MOBILE & MULTI-PLATFORM QA

React Native & Expo, Android (API 36/37) & iOS Fleets, Foldable Viewports, Compact/Flagship Device Extremes.

PERFORMANCE, SECURITY & CLOUD

Supabase Row Level Security (RLS), PostgreSQL, Apache JMeter, Locust Load Testing, WebSockets, Webhooks, SSO Authentication.

DEVOPS, CI/CD & REPORTING

GitLab CI/CD, GitHub Actions, Jenkins, Standalone Zero-Dependency HTML Reports (Inline Base64), Automated Two-Way Test Tracking (Google Sheets API, Trello), Shift-Left Quality Gates.

PROFESSIONAL EXPERIENCE

Senior QA Automation Engineer & AI Test Architect · Skai Technology (Phnom Penh, Cambodia) `06/2026 - PRESENT`

Directing test automation architecture and quality engineering across high-velocity B2B SaaS, mobile native, and enterprise financial platforms.

- **Autonomous Multi-Agent Test Swarms:** Architected a 3-agent specialist swarm (API/Security, Realtime/Webhooks, UI/Workflow) running parallel validation across multi-tenant B2B SaaS platforms, cutting full regression verification cycles by 65%.
- **Zero-Leak Authorization & Security:** Eliminated cross-tenant authorization vulnerabilities by designing validation suites for Supabase Row Level Security (RLS) and additive role pyramids, guaranteeing 100% financial and billing data isolation.
- **Deterministic Web & API Automation:** Automated 52 critical live-data user journeys in Playwright with a 100% pass rate, embedding 98 automated visual assertions and verifying real-time WebSocket sync latencies under 3 seconds.
- **Cross-Platform Mobile Fleet Architecture:** Engineered cross-platform mobile test architecture for React Native/Expo apps across 5 extreme device profiles (foldable viewports, compact Android, and flagship iOS/Android), resolving responsive coordinate clipping before production releases.
- **Mobile Pipeline Speed & AST Bug Isolation:** Implemented Maestro CLI automation reducing smoke test runs from 4 hours to 18 minutes, and diagnosed defects down to the AST and component source code level to accelerate developer fix velocity.
- **Enterprise Load Testing & Standalone Reporting:** Executed Locust and JMeter load benchmarks under high-concurrency SSO traffic, and engineered zero-dependency HTML audit reports with inline Base64 proofs for client and executive sign-off.

PROFESSIONAL EXPERIENCE (CONTINUED)

Quality Assurance Engineer (Sole Lead) · Digital Economy and Business Committee
05/2024 - 01/2026

(Phnom Penh)

Spearheaded quality assurance and release engineering for national digital economy platforms and e-government portals.

- **Institutional QA Infrastructure:** Established the organization's foundational QA infrastructure, authoring Standard Operating Procedures (SOPs), bug classification matrices, and release sign-off gates adopted across all engineering teams.
- **Concurrency & Stress Engineering:** Identified concurrency bottlenecks prior to public rollouts by engineering high-volume stress and load testing suites in Apache JMeter, ensuring server stability under heavy public traffic spikes.
- **AI-Assisted Efficiency:** Accelerated test suite authoring by 50% by designing prompt-engineering pipelines with Claude and ChatGPT to synthesize complex test matrices, edge-case permutations, and API payloads.
- **Zero Critical Escapes:** Maintained a 0% critical bug escape rate across production releases through rigorous manual, API, and cross-browser regression gates.
- **Executive Alignment:** Aligned requirements between developers, product owners, and government executives through structured, data-backed daily release health dashboards.

Quality Assurance Engineer · Infinity Information and App Dev (Phnom Penh)
01/2024 - 04/2024

Delivered manual and API quality validation for client web and mobile applications under rapid two-week sprint cycles.

- **API Contract Verification:** Prevented contract-breaking API changes by building comprehensive Postman test collections, validating JSON schemas, status codes, and latency across all backend services.
- **Comprehensive Test Documentation:** Authored 200+ detailed, reusable test cases covering positive, negative, boundary, and edge-case scenarios accessible to both technical and client stakeholders.
- **Triage Optimization:** Shortened defect triage lifecycles by 30% by submitting reproducible defect reports containing network logs, payload dumps, and environmental parameters.

Quality Assurance Engineer · Techbodia (Phnom Penh)
03/2022 - 09/2022

Developed test automation frameworks and executed performance engineering for enterprise software systems.

- **Automated Regression Testing:** Automated repetitive regression testing by implementing Selenium WebDriver (Python) and Playwright automation frameworks, cutting manual testing effort by 40%.
- **Response Boundary Benchmarking:** Benchmarked system response boundaries using Apache JMeter, identifying database connection pool exhaustion and memory leaks under simulated load conditions.
- **Agile Collaboration:** Integrated automated testing suites into team sprint cadences, participating in Agile ceremonies and executing rapid post-fix verification.

Quality Assurance Engineer / Project Lead · Kirirom Institute of Technology (Kampong Speu)
02/2022 - 10/2022

Led QA operations for computer vision systems and client SDK initiatives, reporting directly to executive leadership.

- **Computer Vision Platforms:** Led QA verification for mission-critical computer vision platforms, including Face Recognition Attendance Systems, License Plate Recognition (LPR) platforms, and client-facing Mobile SDKs.
- **Automated CI/CD Gates:** Built automated CI/CD quality gates using GitLab CI/CD, Jenkins, and Google Cloud Platform (GCP), preventing broken builds from progressing to staging.
- **Executive Milestone Reporting:** Reported weekly milestone velocity directly to the CTO, translating technical defect metrics into clear risk assessments and release decisions.

Learning Management System (LMS) Operator · Kirirom Institute of Technology
02/2022 - 05/2022

- Administered platform workflows, user access controls, and lesson delivery pipelines for faculty and students, ensuring 99.9% LMS availability during exam periods.

EDUCATION & LEADERSHIP

Bachelor of Science in Software Engineering
Kirirom Institute of Technology (KIT)
Kampong Speu, Cambodia

Leadership Program Short Course
University of the Thai Chamber of Commerce (UTCC)
Bangkok, Thailand

ACHIEVEMENTS & LANGUAGES

- **1st Runner-Up** AWS Build on Cambodia Hackathon (2021)
- **Participant** SmartStart & SmartSpark Innovation (2021)
- **Finalist** Techno Innovation Cambodia Hackathon (2021)
- **IELTS 6.5** Academic English Certification (2018)

Khmer: Native | **English:** Advanced (IELTS 6.5) | **French:** Intermediate | **Japanese:** Elementary